

Category Discovery on Instamart: A Gap and an Opportunity

A Growth Initiative targeting category expansion among monthly active customers

CORE ISSUE:

Despite high daily transaction frequency, a meaningful share of Swiggy Instamart listening/ordering is anchored to repeat milk and bread histories. Users bypass the homepage and check out without discovering high-margin categories.

CURRENT GAPS:

- Pre-Checkout Fee Tax: Surcharges on small orders.
- Freshness Distrust: Hard produce return policies.
- Visual Noise: Flashing ads cause navigation anxiety.
- Specification Gaps: Lack of reviews/charts.

Platform	Discovery Feature	2026 AI Move
Instamart	Post-Checkout Add-on	Fee-free sandbox
Zepto	Pass discount widgets	Zepto Pass bundle
Blinkit	Gourmet Specialties	Regional local hubs
BigBasket	BB Daily Subscription	Consolidated deliveries

How the Live Ingestion & Analysis Pipeline Works

The n8n-inspired data cleaning and synthesis workflow: from triggered scrapes to relational insights.

WORKFLOW SEQUENCE:

1. TRIGGER (Webhook) -> 2. SCRAPE (1,000 Multi-Channel reviews) -> 3. MERGE & DEDUPE (Clean duplicates) -> 4. TRANSLATE (Hinglish normalized) -> 5. AI ANALYSIS (Llama sentiment & segment extraction) -> 6. STREAMLIT UI

Curating Ingestion Search Terms:

- *Discovery & Exploration*: search: 'discover', 'explore', 'catalog gap', 'not found'.
- *Recommendation Quality*: search: 'fees', 'surcharge', 'charges', 'promo banner'.
- *Repetition & Fatigue*: search: 'same product', 'reorder', 'history', 'repeat'.

Every Post Classified Into 6 Questions:

1. Why do users struggle to explore new categories?
2. What are the most common frustrations with home banners?
3. What buying behaviors are users trying to achieve?
4. What causes users to repeatedly listen to reorder list history?
5. Which user segments experience different discovery challenges?
6. What unmet needs emerge consistently across discussions?

Discovery Seekers & General Users on Quick Commerce

Segment	Working Professionals / Parents
Age Profile	18-40 years (Gen-Z & millennials)
Locations	Tier 1 & Tier 2 Indian cities
Size	~35M active monthly users
Listening/Cart	75% repeat lists, 94% validation score

Rohan, 29 (Bangalore)	Orders milk daily via history. Delivery fee surcharges on small single add-on items block discovery. Unmet need: Post-checkout append window.
Meera, 42 (Mumbai)	Homemaker. Distrusts fresh produce quality due to past bad batches and wallet-only refunds. Unmet need: Visual expiration/freshness metrics.
Priya, 35 (Pune)	Busy parent. Bypasses diaper catalog on Instamart because the app lacks size charts. Unmet need: Detailed specification sheets.

4 Hypotheses We Assumed Based on AI-Generated Insights

H1: Delivery Fee Tax → Habitual Reordering

Instamart charges full delivery fees on small orders. This blocks users from testing low-risk auxiliary categories as single-item additions.

Tag: Delivery Fee Friction

H3: Visual Overwhelm → Discovery Avoidance

Home screen contains too many flashing banners. Senior or busy cohorts default to reorder history, bypassing discovery tabs entirely.

Tag: UI Clutter & Navigation Anxiety

H2: Freshness Quality → Lack of Catalog Trust

Absence of real-time freshness markers and failure to offer cash refunds on bad produce drives homemakers to local physical markets.

Tag: Quality Trust Deficit

H4: Multi-App Price Comparisons → Basket Splitting

Mixed-language profiles confuse item recommendations. Users split carts across Blinkit and Zepto based on immediate coupon availabilities.

Tag: Platform Fragmentation

Target Segment & Validated Pain Points through Research

Question Tested	Hypothesis	Val %	Verdict
Why struggle to discover?	Recommendations anchored to past histories.	81%	Strongly Validated
What are banner frustrations?	Promo home screen ads do not trigger exploration.	76%	Strongly Validated
What are banner frustrations?	Reorders and search listings are not random.	64%	Validated
What behaviors to achieve?	Users actively seek cross-category discovery.	83%	Strongly Validated
What unmet needs emerge?	A post-checkout append window that waives fees.	76%	Strongly Validated

From Root Cause to Business Case: Why This Is Worth Solving Now

THE ROOT CAUSE:

- Reordering history listings treats familiar repeat items as the safest checkout option, blocking browse-discovery.
- Delivery and packing fees on small orders act as a high-friction tax on new category exploration.
- Niche categories lack detailed specifications inside product pages, driving users to Amazon.

TARGET SEGMENT:

Convenience-First professionals (ages 22-35) representing ~15% of active user base, yielding the highest margin potential.

12-MONTH BUSINESS VALUE (PILOT REGION):

- **Step 1: Segment Sizing:** ~5.2M MACs classified as daily utility buyers (AOV ~\$180).
- **Step 2: Retention Lever:** -0.8pp monthly churn. Saves \$45M in marketing re-acquisition.
- **Step 3: Conversion Lever:** +0.5% conversion of daily grocery buyers into personal care categories.
- **Step 4: Value:** ~\$180 Million ARR lift modeled in pilot region. Zero logistics overhead.

We Scored Three Paths on Reach, Impact, Confidence & Effort: One Winner

Solution Path	Reach	Impact	Confidence	RICE Score
1. Smarter Catalog Filters	Low (1.5M/qtr)	Low (0.5x)	90%	6.8
2. Crate Digger AI Chat Agent	Medium (5M/qtr)	Massive (3x)	50%	12.5
3. Post-Checkout Append Window	High (20M/qtr)	High (2.0x)	80%	106.6

Post-Checkout Append Window wins because it sits on a screen every listener already sees, one click is enough to act on it, and it reuses the assigned rider already traveling to the location.

User Flow Diagram & Backend System Architecture

USER FLOW DIAGRAM (PILOT SESSION):

[Order Placed] -> [App checks: is it grocery/milk?]
-> YES

[Active 3-Min Timer Banner Starts] -> [User views
recommended items] -> [User clicks Add-to-Order]

[System runs validation: is timer active?] -> YES

[Waives delivery fee to ■0] -> [Appends item in
active SQLite DB session] -> [Updates total bill] ->
[Pushes updated manifest to Dark Store Packer]

SYSTEM ARCHITECTURE (PHASE 0 - PHASE 4):

- **Phase 4: UI/Orchestration Layer:** Mode toggle A/B/C checkout sandbox. Timer displays, callback handlers.
- **Phase 3: Reasoning Layer:** FastAPI gateway API route `/api/opportunities` serves prioritized opportunities.
- **Phase 2: Matchers:** Cross-category engine pairs milk orders with beauty (face wash) and organic produce (avocados).
- **Phase 1: Storage Layer:** SQLite3 (`reviews.db`) stores user histories, active checkouts, and ICE opportunity roadmaps.

Why Post-Checkout Sandbox, and Not Traditional Recommenders?

THE GAP: Why Banners Fail

- Home screen ads are treated as spam (banner blindness).
- Collaborative filtering recommends items the user has already bought, locking them in loops.
- Pre-checkout widgets trigger fee warnings, creating cart abandonments.

THE UNLOCK: What the MVP Unlocks

- Leverages active transaction momentum immediately post-checkout.
- Waives delivery fee for 3 mins as rider cost is already paid.
- Recommends high-margin exploratory categories (beauty, pets) based on daily context.

THE OUTCOME: How UX Transforms

- Cart additions shift from high-consideration friction to low-friction taps.
- Users explore auxiliary categories without opening competitor apps.
- Instamart transitions from a commodity milk-run channel to a multi-category shopping center.

The reasoning is simple: a traditional pre-checkout banner increases cart abandonment; but a post-checkout timer leverages warm payment intent to trigger zero-fee discovery when the user's focus is at 100%.

Go To Market Strategy & Growth Guardrail Metrics

PHASED ROLLOUT ROADMAP:

- **Phase 1 (Weeks 1-4): Controlled Pilot:** Launch in Bangalore east dark stores. Limit to active grocery buyers with 5+ reorder history. Target: >25% completion, 0% rider delays.
- **Phase 2 (Weeks 5-10): Targeted Expansion:** Expand to top-10 dark stores. A/B test inline countdown timer banner vs bottom-sheet slide. Target: >20% toggle, >35% retention.
- **Phase 3 (Weeks 11-16): Full Rollout:** Enable for all active users. Link to Swiggy One membership to drive subscription upsell.

Risk Event	Impact	Mitigation Strategy
Rider delay due to late additions	Med/High	Disable append if packer scanned order barcode.
User abuses free fee for bulk items	High/Med	Limit add-ons to light items (<2kg) matching bike capacity.
Gourmet items are out of stock	Med/Low	Sync add-on shelf with dark store live inventory counts.